

and then summing the resulting products:

$$\begin{bmatrix} -1 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_1 + 5x_2 - 3x_3 \end{bmatrix}$$

Likewise, the third entry in Ax can be calculated from the third row of A and the entries in x . $\square$

Row-Vector Rule for Computing Ax

If the product Ax is defined, then the i th entry in Ax is the sum of the products of corresponding entries from row i of A and from ^{the} vector x .

Example 5:

$$a) \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 3 + (-1) \cdot 7 \\ 0 \cdot 4 + (-5) \cdot 3 + 3 \cdot 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$b) \begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \cdot 4 + (-3) \cdot 7 \\ 8 \cdot 4 + 0 \cdot 7 \\ -5 \cdot 4 + 2 \cdot 7 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix}$$

$$c) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} 1 \cdot r + 0 \cdot s + 0 \cdot t \\ 0 \cdot r + 1 \cdot s + 0 \cdot t \\ 0 \cdot r + 0 \cdot s + 1 \cdot t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix} \quad \square$$

- By definition, the matrix in Example 5(c) with 1's on the diagonal and 0's elsewhere is called an identity matrix and is denoted by I .
- The calculation in part (c) shows that $Ix = x$ for every x in $\mathbb{R}^3$.
- In $\mathbb{R}^n$, the $n \times n$ identity matrix. $Ix = x$ for every x in $\mathbb{R}^n$.

Properties of the Matrix-Vector Product Ax

Theorem 5: If A is an $m \times n$ matrix, u and v are vectors in $\mathbb{R}^n$, and c is a scalar, then:

- $A(u+v) = Au + Av$
- $A(cu) = c(Au)$